Part One: Foundations, Regression, and Generalized Linear Models

An Introduction to the Chapters and How They Fit Together

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1 What Part One Offers

Quantitative Methods for Social and Behavioral Research develops the statistical understanding needed to read empirical studies, interpret their results, and conduct quantitative research. Part One is where the book begins: fourteen numbered chapters that start with what a study asks and how it produces its evidence, and end with regression models for a range of outcomes. It is written for postgraduate students and applied researchers. A mean, an interval, or a regression coefficient becomes useful when a reader understands what it describes, how it was obtained, and which question it helps answer.

Some introductory familiarity with statistics and algebra is useful, and little else is assumed. Each chapter explains the notation it uses and develops its calculations in worked steps, so the algebra arrives with the idea that needs it. Calculus and proof-based mathematical statistics are not prerequisites. Chapters 1–4 can be read without running R or SPSS: Chapter 1 introduces the research journey, and Chapters 2–4 teach statistics directly, through explanations, figures, and worked calculations. Computation joins the statistics in Chapters 5 and 6.

Results are rounded to four decimals from full-precision computation, with trailing zeros dropped (3.1 rather than 3.1000), unless a source's own precision applies.

2 How the Chapters Build on Each Other

Part One contains fourteen chapters. The first nine establish a route from questions and records to regression. Chapter 1, *The Quantitative Research Journey*, orients the reader in questions, evidence, design, and interpretation. Chapter 2, *Data, Distributions, and Variation*, develops records, distributions, and variation across repeated samples. Chapter 3, *Estimates, Uncertainty, and Statistical Tests*, develops estimation, intervals, and tests. Chapter 4, *Group Comparisons with Analysis of Variance*, develops comparisons among several groups. Chapter 5, *Description and Basic Inference with R*, and Chapter 6, *Group Comparisons and Statistical Modelling with R*, make those foundations executable. Chapters 7–9 then develop regression from one predictor to the regression toolbox: Chapter 7, *Simple Linear Regression*; Chapter 8, *Multiple Regression I: Conditioning, Checks, and Coordinated Questions*; and Chapter 9, *Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty*.

Chapters 10–14 extend the regression framework to conditional relationships, noncontinuous outcomes and prediction, then return to the design of a study and the limits of its claims. Interaction changes how a relationship varies across conditions; generalized linear models change the outcome model; validation asks how a fitted rule performs on new cases; and causal design asks which comparisons can support an intervention claim.

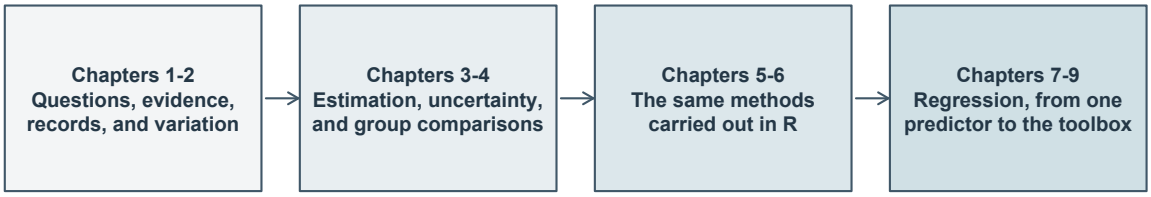


Figure 1: The foundations of Part One, from questions and records to inference, computation, and regression in Chapters 1–9.

Table 1 states what each chapter is responsible for and what it builds on. Chapters 1–9 are available on the website. Chapters 10–14 are not yet available; their rows outline the continuation of the curriculum.

Table 1
What each chapter is responsible for.

Chapter	What it is responsible for	What it builds on
1. The Quantitative Research Journey	Turning a concern into a question with a stated target quantity; how a study produces its evidence; how comparison, timing, sampling, and assignment support descriptive, associational, predictive, and causal answers; reading a reported result as a bounded claim	The starting point of the part
2. Data, Distributions, and Variation	Reading records, identifiers, codes, and denominators; sampling schemes; distributions, their skew and tail weight, centre, spread, and groups; relationships between two variables; how a summary varies across repeated samples	Reads the dataset behind Chapter 1's account of evidence and applies its design reasoning to an interpretation
3. Estimates, Uncertainty, and Statistical Tests	The target quantity, the rule that estimates it, and the value obtained; standard errors for a mean and a difference, including the pooled standard error, and the conditions behind them; confidence intervals and confidence level; hypothesis tests, p-values, and significance; the one-sample, paired, equal-variance, and Welch t tests; the chi-square test of association and tables of three or more categorical variables; magnitude, error rates, power, and choices made after seeing the results	Turns Chapter 2's sampling distributions and standard error into inference from one observed sample
4. Group Comparisons with Analysis of Variance	Group-mean models and their comparison by the F test; one-way and two-way analysis of variance; sums of squares, degrees of freedom, and mean squares; contrasts and comparison families; interaction and simple effects, extended to a worked three-way table and the structure of four-way designs	Applies Chapter 2's description of group variation and Chapter 3's inference to comparisons among several groups

Chapter	What it is responsible for	What it builds on
5. Description and Basic Inference with R	Preparing records in base R with explicit missing values; distributions, graphics and probability calculations; relationships, covariance and correlation matrices, and one-control partial correlation; simulated sampling variation, bias and standard error; mean intervals and their repeated-study coverage, z and t tests, and selected correlation inference; a short reproducible report	The computational route through Chapter 2's order of distributions, relationships and sampling, followed by Chapter 3's mean inference
6. Group Comparisons and Statistical Modelling with R	Group-mean and factorial models in R, with the F test comparing named models; graphical comparisons with ggplot2; contrasts and comparison families, including a declared contrast and the protection needed if it had been selected after inspection; proportions and counts, with their effect scales; numerical optimization through squared loss and a scalar likelihood; simulation, precision and power	Implements Chapter 4's model comparisons and extends Chapter 3's categorical and planning lessons on Chapter 5's computational foundation
7. Simple Linear Regression	The model for a conditional mean and its error variation, first through a known process and then occupation records; least-squares estimates and slope inference; repeated-study uncertainty; mean-response and new-case intervals; diagnostics; regression toward the mean, its paradox and its causal misreading, with Galton's heights and repeated exams	Extends Chapter 4's group-mean model to a line, using Chapters 5–6's covariance, inference, simulation and objective-function work
8. Multiple Regression I: Conditioning, Checks, and Coordinated Questions	The multiple-regression model and partial slopes through a known process; estimation and inference on the same occupation records, with the overall F test; checks, coefficient covariance, fitted profiles and standardization; a separate held-out prediction question; three coordinated answers in one study	Extends Chapter 7's model and line with Chapter 5's partial correlation and Chapter 4's model comparisons, implemented in R in Chapter 6

Chapter	What it is responsible for	What it builds on
9. Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty	Changes to the mean part: indicators and contrasts, covariance adjustment, nested and non-nested model comparisons, and changed forms and scales; changes to uncertainty: HC3, resampling, and a randomized study with school clusters	Extends Chapter 8's model, connecting each method to the part it changes and the conditions it needs
10. Interaction and Moderation	Product terms, conditional slopes, higher-order interaction and moderated moderation, comparisons within joint support, and bootstrap uncertainty	Extends Chapters 8 and 9's regression models, contrasts and resampling to relationships that vary with another variable
11. Binary Logistic Regression	The Bernoulli likelihood, the logit link, probability and odds scales, fitting, comparison and checks for a binary outcome	Changes Chapter 8's outcome model while retaining its conditional-question discipline and Chapter 6's likelihood foundation
12. Extending the GLM: Nominal, Ordinal, and Count Outcomes	Outcome classification, generalized and cumulative logits, counts and rates, overdispersion, zero-inflated and hurdle models, and model checks	Extends Chapter 11's response-distribution, link and linear-predictor framework
13. Prediction, Cross-Validation, and Regularization	Held-out error, leakage, penalized fitting, classification performance and changes in the deployment population	Develops Chapter 8's separate prediction question with the regression and classification models of Chapters 9–12
14. Regression Pitfalls and Causal Design	Adjustment structures, reversals, selection, measurement, dependence and the design support for a bounded claim	Re-examines the preceding models through Chapter 1's questions, evidence and interpretation, including what adjustment alone cannot supply

Each idea has one principal home, and later chapters use it rather than teach it again. Chapter 1 previews uncertainty while reading a study; Chapter 2 develops sampling distributions and the standard error through repeated samples; Chapter 3 develops their inferential use and states the conditions they require. Chapter 2 explains the descriptive ingredients of group

variation; Chapter 4 develops the full analysis-of-variance decomposition and its inferential use, comparing one common mean with group means and an additive model with a model that includes interaction. Where a later chapter needs an earlier idea, it recalls it briefly and then puts it to work in the new setting.

3 How to Read a Chapter and Use the Concept Reference

The numbered chapters share an arrangement. A **Chapter overview** box opens the chapter with the route it follows and what a reader will be able to do afterwards. Numbered sections develop the material in order. A closing **Reading Bridge** brings the principal ideas together and connects them to the next chapter, and a selective **Key terms** table gives a compact reference to the concepts explained principally there. Unnumbered material follows the numbered sections: exercises for working with the ideas, appendices holding fuller derivations and specialized extensions, and the references. The arrangement keeps the main explanation distinct from fuller derivations and specialized extensions.

There are two sensible routes through the chapters. The first is the chapter sequence. The second pairs statistics with computation: Chapters 2 and 3 with Chapter 5, then Chapters 3 and 4 with Chapter 6. Either way, follow a method from its data and question through its calculation to its interpretation; a result separated from the question it answers is difficult to use.

The **Part One Concept Reference** is published beside this introduction. It gathers selected concepts from Chapters 1–14 into nine tables, arranged by concept group and then by term, with each term's meaning and the place where the book explains it principally. A further table lists the procedures of Chapters 3 and 4 with what each estimates and the conditions it relies on, and a closing table records the notation those chapters use. It is a place to look something up while reading rather than a substitute for reading, because first-use explanations and worked examples stay in the chapters themselves.

Readers who would like to run R without installing anything can use the public browser practice pages that accompany Chapters 3 and 4.

4 What Comes Later

Three argument chapters extend the argument-based approach. Argument Chapter A, *How to Read a Quantitative Study as an Argument*, develops the reading, evaluation, and construction of one question-level argument. Argument Chapter B, *Extending the Argument-Based Approach to Quantitative Studies*, develops arguments with several statistical strands and studies with coordinated questions. Argument Chapter C concerns arguments involving latent variables or an additional measurement layer, drawing on Part Two. The argument chapters are optional, and no numbered chapter depends on them.

Argument Chapter C is reserved for that further development.

Part Two of the book turns to latent-variable modeling, where the variable of primary interest is one that no column of the data set holds. Its twelve chapters proceed from factor analysis and measurement models to structural systems, causal identification, indirect and conditional relationships, group comparisons, incomplete data, change and study planning.